

Conjecture 00000008330: $(-2, -1) \cup (1, 2)$ is not the list of gaps of $C - C$

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1 The conjecture

English. Definition: The arithmetic sums of Cantor sets: the higher-dimensional products beyond the known $C + C = [0, 2]$ for the middle-thirds Cantor set C . Conjecture: The product $C \times C$ has the critical dimension sum $4/3$ (a critical sum law); the complete list of gaps of the difference set $C - C$ (complement intervals) is the double gap $(-2, -1) \cup (1, 2)$ (a double gap law); and the fullness criterion for arithmetic sums of digit-restricted sets (base- b digit-restricted sets) is critical at the Newhouse threshold $\dim_1 + \dim_2 = b$ (a Newhouse fullness threshold law).

Chinese text. The Chinese version has the same three clauses; the second reads (translated) “the complete list of the holes of the difference set $C - C$ (the intervals of the complement of the difference set) is the double hole $(-2, -1) \cup (1, 2)$ ”.

2 Reading

The conjecture is a conjunction of three clauses. We refute the second clause (the “double gap law”); this refutes the conjunction. We do not use or assess the first clause (dimension sum $4/3$) or the third clause (Newhouse threshold).

Definition 1. C is the middle-thirds Cantor set: $C = \bigcap_{n \geq 0} C_n$, where $C_0 = [0, 1]$ and $C_{n+1} = \frac{1}{3}C_n \cup \frac{1}{3}(2 + C_n)$. This is Mathlib’s `cantorSet` (file `Mathlib/Topology/Instances/CantorSet.lean`). The difference set is $C - C = \{x - y : x, y \in C\}$ (pointwise subtraction, `cantorSet - cantorSet`).

The text identifies the gaps with “complement intervals”. We refute the clause under each of the following four readings of “the complete list of gaps of $C - C$ is $(-2, -1) \cup (1, 2)$ ”:

- (R1) *complement reading*: $\mathbb{R} \setminus (C - C) = (-2, -1) \cup (1, 2)$;
- (R2) *standard gaps*: a gap of a set $S \subseteq \mathbb{R}$ is a bounded connected component of $\mathbb{R} \setminus S$ (Lean: `IsGap`); the clause says in particular that $(1, 2)$ and $(-2, -1)$ are gaps of $C - C$;
- (R3) *ambient interval* $[-2, 2]$: $(1, 2)$ is a connected component of $[-2, 2] \setminus (C - C)$;
- (R4) *gaps inside the hull*: gaps are the components of $[\inf(C - C), \sup(C - C)] \setminus (C - C)$, so each gap lies inside $[\inf(C - C), \sup(C - C)]$.

Every reading in which the listed intervals are complementary intervals of $C - C$ requires $(1, 2)$ to be a component of the complement of $C - C$ in some ambient set; (R1)–(R4) are the ambient choices $\mathbb{R}$, $[-2, 2]$ and the hull. A reading that takes the complement inside some other ambient set (for example $(1, 2)$ itself) is not covered.

3 Proof

Lemma 2. $C \subseteq [0, 1]$ and $0, 1 \in C$. Hence $C - C \subseteq [-1, 1]$, $\pm 1 \in C - C$, $\sup(C - C) = 1$ and $\inf(C - C) = -1$.

Proof. $C \subseteq C_0 = [0, 1]$, and $0 \in C$ (Mathlib `cantorSet_subset_unitInterval`, `zero_mem_cantorSet`). By induction $1 \in C_n$ for all n , since $1 = (2 + 1)/3$; so $1 \in C$. If $x, y \in [0, 1]$ then $-1 \leq x - y \leq 1$. Finally $1 = 1 - 0$ and $-1 = 0 - 1$. $\square$

Theorem 3. (R1) $\mathbb{R} \setminus (C - C) \neq (-2, -1) \cup (1, 2)$.

(R2) For every x , the connected component of x in $\mathbb{R} \setminus (C - C)$ is neither $(1, 2)$ nor $(-2, -1)$. In particular neither $(1, 2)$ nor $(-2, -1)$ is a gap of $C - C$.

(R3) For every x , the connected component of x in $[-2, 2] \setminus (C - C)$ is not $(1, 2)$.

(R4) $(1, 2) \not\subseteq [\inf(C - C), \sup(C - C)] = [-1, 1]$.

Proof. By Lemma 2, $(1, \infty)$ and $(-\infty, -1)$ lie in $\mathbb{R} \setminus (C - C)$. (R1): $3 \notin C - C$ but $3 \notin (-2, -1) \cup (1, 2)$. (R2): suppose the component of x equals $(1, 2)$. It is nonempty, so x lies in the complement, and then x lies in its own component, so $x \in (1, 2)$. The interval $(1, \infty)$ is connected, contains x and lies in the complement, so it lies in the component of x . Then $3 \in (1, 2)$, a contradiction. The case $(-2, -1)$ is symmetric (use $(-\infty, -1)$ and -3). (R3): in the same way, $x \in (1, 2)$, and $(1, 2]$ is connected, contains x and lies in $[-2, 2] \setminus (C - C)$. So 2 lies in the component, but $2 \notin (1, 2)$. (R4): $3/2 \in (1, 2)$ but $3/2 > 1 = \sup(C - C)$. $\square$

Remark 4. The proof uses only $C \subseteq [0, 1]$ and $0, 1 \in C$. In fact $C - C = [-1, 1]$, because $C - C = C + C - 1$ (C is symmetric about $1/2$), but neither this nor $C + C = [0, 2]$ is used or formalized.

4 Lean formalization

File `lean/Conjecture8330/Basic.lean` (Lean 4, Mathlib v4.33.1; its only import is `Mathlib`). It uses Mathlib's `cantorSet` and defines `IsGap S I := ($\exists x \in S^c$, connectedComponentIn S^c x = I) $\wedge$ I bounded. Theorem names: C8330.sub_subset_Icc, C8330.one_mem_cantorSet, C8330.sSup_sub, C8330.sInf_sub (Lemma 2); C8330.compl_ne_double_gap (R1); C8330.Ioo_one_two_not_component, C8330.Ioo_neg_two_neg_one_not_component (R2); C8330.Ioo_one_two_not_component_in_Icc (R3); C8330.Ioo_one_two_not_subset_hull (R4). The main theorem combines them:`

$$\begin{aligned} & \neg \text{IsGap } (C - C) (1, 2) \wedge \neg \text{IsGap } (C - C) (-2, -1) \wedge (C - C)^c \neq (-2, -1) \cup (1, 2) \wedge \\ & (\forall x, \text{connectedComponentIn } ([-2, 2] \setminus (C - C)) x \neq (1, 2)) \wedge \\ & \neg((1, 2) \subseteq [\inf(C - C), \sup(C - C)]), \end{aligned}$$

where C is `cantorSet`, (a, b) is `Ioo a b`, $[a, b]$ is `Icc a b`, and $\inf, \sup$ are `sInf`, `sSup`.

Axiom audit. `#print axioms C8330.double_gap_law_false` reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.